Research on the Design and Implementation of a Mesh Network Communication System for Fire and Hazardous Gas Detection and Response

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***Abstract*—This paper proposes an evacuation route guidance system based on a mesh network that detects hazardous factors such as fire and toxic gases in industrial sites and buildings and quickly guides evacuation routes in urgent situations at disaster sites. The proposed evacuation route guidance system detects hazardous air conditions, such as fire and gases such as carbon monoxide and formaldehyde, through sensors and uses 5G and Wi-Fi mesh networks to offer real-time risk information and evacuation routes to users and managers.**

Keywords—Smart factory; Wireless Mesh Network; Fire Management; Fire Information Collection; 5G communication

I. Introduction

According to the full enforcement of the Serious Accident Prevention Act, interest in industrial safety has been increasing, with various studies conducted on disaster prevention technologies using smart factory technologies applied in different ways in manufacturing sites. However, small and medium-sized enterprises, which constitute a large portion of industrial accidents, encounter difficulties in preventing industrial accidents through the application of advanced ICT technologies owing to practical problems. Furthermore, one of the leading causes of danger at disaster and accident sites is the occurrence of hazardous gases due to gas leaks and fire combustion. Although demand for new fire-fighting facilities is increasing as industrial complexes become smarter and more complex, and the architectural environment changes, operational and evacuation support capabilities of fire-fighting products and technologies during fire and gas leak incidents are not satisfying expectations.

In today's society, residential and industrial environments are transforming into more high-rise and dense forms, resulting in inadequate initial response and evacuation systems in the event of accidents. To reduce human casualties caused by fires, a smart fire and hazardous gas detection and warning system that is equipped with rapid and accurate initial detection technology and is capable of guiding the optimal evacuation routes should be developed [1–3].

In addition to the existing gas leak and fire alarm functions, an integrated solution that can prevent human casualties by incorporating an LED-based evacuation alert system to support the quick evacuation of workers from the fire site is required. At the time of a fire in particular, P-type receivers connected to multiple sensors within a certain range must be able to exchange information about the location of the fire with other P-type receivers in a bidirectional, real-time manner, thus necessitating mesh network construction technology using Wi-Fi for sensor devices [4].

This study aimed to develop a customized monitoring and evacuation route guidance system for detecting fire and hazardous gas leaks, thereby preventing industrial accidents. The objective is to detect hazards such as electrical leakage, overcurrent, fires due to high temperatures, and the emission of hazardous substances through gas and temperature sensors and to investigate a fast evacuation route guidance and safety management system based on 5G and Wi-Fi communication.

II. Proposal of the Design of a Mesh Network Communication System for Fire and Hazardous Gas Detection and Response

A fire and hazardous gas monitoring system using 5G communication and Wi-Fi (IEEE 802.11aX) modules detects hazardous gases indoors or outdoors in real-time using multiple gas and temperature sensors, analyzes these data to generate concentration data for each type of gas detected by the sensors, and transmits this information wirelessly or by wire to enable users to monitor the situation and manage safety accidents in real time.

Beyond the basic functions of standalone detectors commonly used in industrial sites, the design enhances reliability by acquiring accurate disaster information through the functions of P-type receivers and external Wi-Fi and 5G modules for real-time data measurement (hydrogen sulfide

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gas, other hazardous gases, carbon monoxide) and data validation (data errors, sensor trash data).

The proposed P-type receiver is a 5G and Wi-Fi-based receiver that can be interconnected with 10 detectors circuits, designed to power the detectors and enable basic operations. Furthermore, automatic fire detection facilities are designed to detect early signs of fire such as heat, smoke, or flames automatically, issuing alarms to discover fires early, thereby enabling early notification, initial suppression, and early evacuation. In the event of a fire, it operates automatically through detectors installed in the ceiling or a boarded ceiling, or it operates by manually triggering a transmitter to input the fire signal to the receiver. When the fire signal is received, bells and visual alarms are activated, and if installed, fire-fighting and evacuation facilities automatically operate based on the building's capacity, total floor area, and number of floors.

The power board of the P-type receiver has been modified from the output voltage setting used in existing residential kitchen automatic fire extinguishing devices, and the transmitted power line is not a single sinusoidal AC voltage but includes unwanted high-frequency voltages, posing a risk of circuit damage. Therefore, to counteract these noise signals and surge voltages, the power circuit has been designed using TNR, high-voltage ceramics, and fuses.

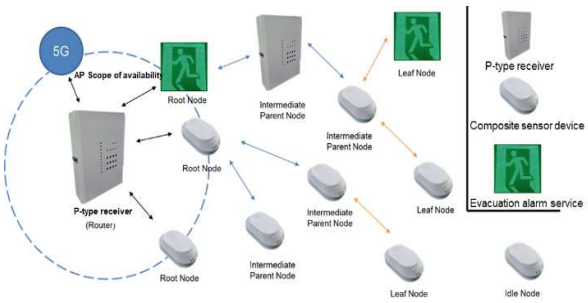

Fig. 1. Overview of the mesh network-based communication system for fire and hazardous gas detection and response

The complex sensor device circuit was designed to provide a power section with a 5 V 2 A output through a DC–DC converter circuit, which is powered by a 24 V input from the terminal board, which in turn powers the 3.3 V 1 A output MCU and power circuits for Wi-Fi and 5G communication modules. The Wi-Fi and 5G module communication transmits and receives data using the wireless communication function of the P-type receiver. The 5G to Wi-Fi6 CPE hardware prototype was designed with an M.2 form factor to insert Ethernet PHY and 5G modules. This distributes network IP over a wired connection in a 5G communication environment and supports TCP/IP communication for client devices. The Wi-Fi mesh system operates stably through mutual cooperation and data transmission among various nodes, with central management and relay [5–7].

Each node in the complex sensor device functions at an equal level of the stack, maintaining an egalitarian relationship without level influence. Nodes are arbitrarily classified into root, intermediate parent, and leaf nodes. The root node serves as the top node of the network, responsible for the central management and control of the entire network. The intermediate parent node functions as a mid-level node, enhancing network scalability and efficiency through data relay and routing. The leaf node operates at the end level, serving as the source or destination of data, and is responsible for collecting and controlling actual sensor data.

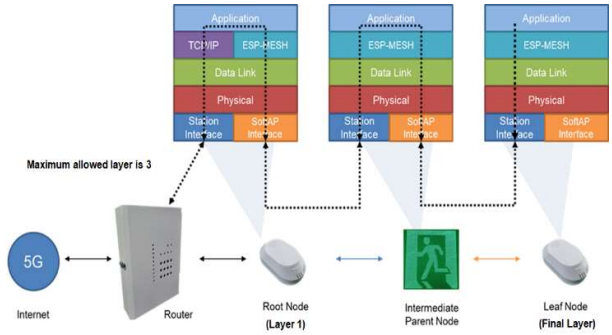

Fig. 2. Example of data stream configuration in a mesh network-based communication system

The P-type receiver controls relay through peripheral device control functions, managing the power supply status of various peripheral devices. With the LEDs on the MCU board and terminal board, users can easily check the power status of peripheral devices. Therefore, the P-type receiver can efficiently manage and monitor the power supply of peripheral devices.

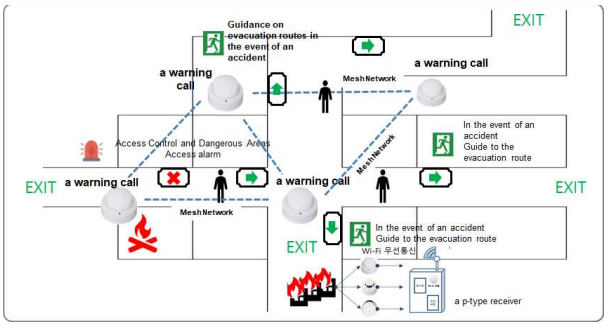

Fig. 3. Example of an evacuation route guidance scenario in the event of a fire incident

In the entire building, corridors and rooms (nodes) are assigned weights, and the shortest path is determined according to the tree structure based on these weights. Weights are determined based on the goods stored, fire/hazardous gas detection, and weights set by the manager, ranging from 1 to 5, where 1 represents no restrictions and 5 indicates fire/hazardous gas detection, with 2 to 4 classified according to the stored goods or arbitrarily set by the manager. If the weight is 9, it is automatically excluded from the evacuation route guidance algorithm; otherwise, lower numbers are prioritized to construct the shortest path to the exit.

III. Implementation and Experiments

The mesh network-based P-type receiver uses a separate connector for high-speed communication interfaces with 5G and Wi-Fi modules. The control part adjusts the operation of the terminal and manages data communication for swift and accurate information exchange. We implemented Wi-Fi mesh functionality using the ESP32 microcontroller module. It was developed using the same stack as the Wi-Fi mesh applied to sensor devices utilizing Winsen's ME2-CO. Through this, it communicates with other P-type receivers or ESP32-based

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devices, exchanging evacuation route information. The ESP32 displays the received evacuation route information on the LED display in the form of arrows or text, providing users with information on the direction of evacuation.

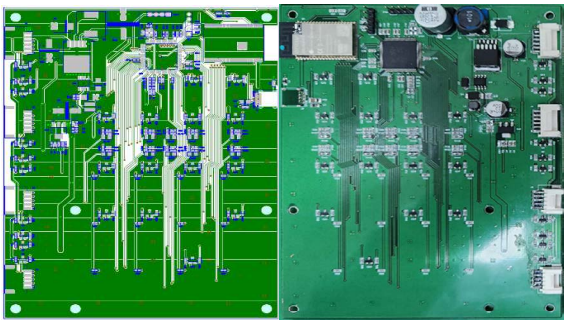

Fig. 4. PCB design and prototype photo of the P-type receiver

The firmware configuration of the mesh network-based P-type receiver measures analog voltage values for carbon monoxide sensors, VOCs sensors, photoelectric smoke detectors, and gas sensors, and determines detection based on the set sensing thresholds. The mesh network begins with the process of creating a network, which involves registering one or more nodes and forming connections between them. In TCP socket connection, if the connection fails, LogW: ret is indicated as –1. (If the TCP connection fails, ret returns –1 and repeats until a connection is established.) Success in the TCP socket connection starts TCP communication without a separate success notification log. The configuration information (such as the current channel in use and the addresses of the nodes connected to it) of the mesh system is updated periodically.

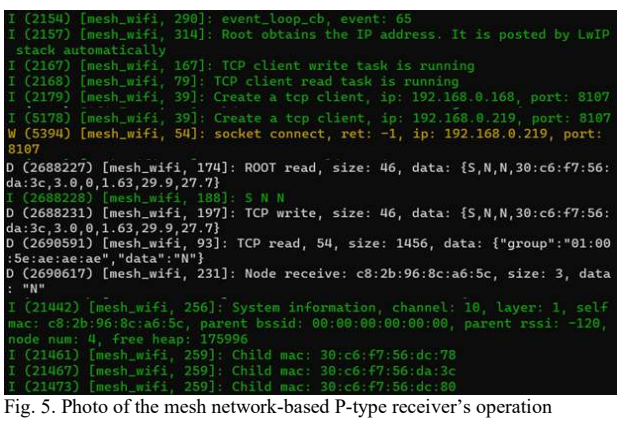

Fig. 5. Photo of the mesh network-based P-type receiver's operation

IV. Conclusion

In this study, a safety management system was designed and implemented to prevent industrial accidents in real-time through 5G and Wi-Fi wireless networks by detecting fires and hazardous gas leaks using complex sensors. In public safety, the wireless mesh network enables the provision of wireless networks for firefighting, police, and emergency response scenarios. It can provide effective public safety communication by offering various services such as video surveillance, emergency responder tracking, voice, and data communication.

This equipment is essential for minimizing human and property damage by acquiring ICT-based status data for fire detection in various facilities. It enables early detection and notification in the event of industrial accidents, alerting relevant parties quickly. Therefore, a significant increase in demand is anticipated.

In the industrial field, the market is gradually expanding for wireless technologies such as Bluetooth and Zigbee, primarily applying WLAN (wireless industrial Ethernet) to address the limitations of wired connections. Research has been conducted actively on 5G-TSN (time-sensitive networking) technology, which enables integrated control of data streams, mission-critical traffic, and non-mission critical traffic, and latency-free, real-time, reliable, and secure performance. Various studies are planned on fusing 5G technology and industrial networks, which can satisfy all the requirements of the industrial field, not only improving the flexibility of factory automation and operational management but also enabling reliability and low latency in emergency scenarios.

Acknowledgment

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